

Generation Interconnection System Impact Study Report

For Queue Project Q14-0360

Haleyville Ts 100 kV

May 26, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0360. The Interconnection Customer proposes a 265.5 MW wind facility interconnecting at the Haleyville Ts 100 kV substation in the Indiana area. The request was received on August 05, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 176 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$14,054,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Auburn-Haleyville Ts 100.0 kV: 241.1% loading (119 MVA) following the outage of the Minneapolis City Of-Wallenpaupack 500.0 kV; DFAX = 100.0%.

The Auburn-Noble 100.0 kV Line is overloaded to 215.4% of its 127 MVA emergency rating for the loss of the Minneapolis City Of-Wallenpaupack 500.0 kV. The project contributes 100.0% DFAX (265.5 MW) to this overload.

The Allings Crossing-Noble 100.0 kV is overloaded to 193.0% of its 133 MVA emergency rating for the loss of the Minneapolis City Of Substation-Wallenpaupack 500.0 kV. The project contributes 100.0% DFAX (265.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Auburn-Haleyville Ts 100.0 kV	Rebuild line with higher-capacity conductor	\$6,321,000	86% (\$5,463,000)
Auburn-Noble 100.0 kV Line	Rebuild line with higher-capacity conductor	\$7,368,000	86% (\$6,368,000)
Allings Crossing-Noble 100.0 kV	Rebuild line with higher-capacity conductor	\$2,573,000	86% (\$2,223,000)
	Total allocated to project		\$14,054,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.